ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 30 cm	$r^2 = OP^2 - PT^2 = 34^2 - 16^2 = 900$; $r = 30$ cm
2	**(d)** 92 cm	Perimeter $= PA + AO + OB + BP = 2(16) + 2(30)$; Perimeter $= 92$ cm
3	**(a)** 360 cm²	Area $= 2 \times \frac{1}{2} \times 40 \times 9$; Area $= 360$ cm^2
4	**(a)** 96 cm	Half-chord $= \sqrt{52^2 - 20^2} = 48$ cm; Chord $= 2 \times 48 = 96$ cm
5	**(b)** 5 cm	$AF = AB - r = 11 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(11 - r) + (60 - r) = 61$ gives $r = 5$ cm
6	**(a)** 43 cm	$8x + 3 = 5x + 18$ gives $x = 5$; $QA = 8(5) + 3 = 43$ cm
7	**(a)** 110°	$\angle APB = 180^\circ - 35^\circ - 35^\circ$; $\angle APB = 110^\circ$
8	**(a)** 13 cm	$OP^2 = 144 + 25 = 169$
9	**(d)** 28 cm	$r^2 = OP^2 - PT^2 = 35^2 - 21^2 = 784$; $r = 28$ cm
10	**(a)** 98 cm	Perimeter $= PA + AO + OB + BP = 2(21) + 2(28)$; Perimeter $= 98$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$PA = PB$, so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB$; $\angle APB = 180^\circ - 30^\circ - 30^\circ$ (1); $\angle APB = 120^\circ$ (1).	1+1
12	$PA = PB$, so $\angle PAB = \frac{180^\circ - 36^\circ}{2} = 72^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 72^\circ = 18^\circ$ (1).	1+1
13	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 52^2 - 48^2 = 400$ (1); $PT = 20$ cm (1).	1+1
14	$PA = PB$, so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB$; $\angle APB = 180^\circ - 55^\circ - 55^\circ$ (1); $\angle APB = 70^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tangents from an external point are equal: $QA = QB$ (1); $9x + 4 = 5x + 20$ gives $x = 4$ (1); $QA = 9(4) + 4 = 40$ cm (1).	1+1+1
16	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $18 + 12 = AD + 15$ (1); $AD = 15$ cm (1).	1+1+1
17	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 30 \times 16$ (1); Area $= 480$ cm^2 (1).	1+1+1
	OR: Tangents from an external point are equal: $QA = QB$ (1); $7x + 2 = 4x + 14$ gives $x = 4$ (1); $QA = 7(4) + 2 = 30$ cm (1).	1+1+1
18	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $21 + 14 = AD + 17$ (1); $AD = 18$ cm (1).	1+1+1
	OR: $\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 28 \times 21$ (1); Area $= 588$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{52^2 - 20^2} = 48$ cm; perimeter $= 2(48 + 20)$ (2); Perimeter 136 cm; area $= 2 \times \frac{1}{2} \times 20 \times 48 = 960$ cm^2 (1).	2+2+1
	OR: $OA \perp PA$ and $OB \perp PB$, so $\angle OAP = \angle OBP = 90^\circ$ (2); In quadrilateral $OAPB$ the angle sum is 360° (1); $\angle APB + \angle AOB = 360^\circ - 90^\circ - 90^\circ = 180^\circ$ (2).	2+1+2
20	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{53^2 - 28^2} = 45$ cm; perimeter $= 2(45 + 28)$ (2); Perimeter 146 cm; area $= 2 \times \frac{1}{2} \times 28 \times 45 = 1260$ cm^2 (1).	2+2+1
	OR: $PA = PB$ and $OA = OB$, so P and O both lie on the perpendicular bisector of AB (points equidistant from A and B) (2); Alternatively, let OP meet AB at M; $\triangle PAM \cong \triangle PBM$ (SAS, since $\angle APM = \angle BPM$) (2); So $AM = MB$ and $\angle PMA = \angle PMB = 90^\circ$ (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90^\circ$; the correct sum is 180°.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 972 cm²	Area $= 2 \times \frac{1}{2} \times 36 \times 27$; Area $= 972$ cm^2
2	**(d)** 90 cm	Half-chord $= \sqrt{53^2 - 28^2} = 45$ cm; Chord $= 2 \times 45 = 90$ cm
3	**(a)** 7 cm	$AF = AB - r = 16 - r$, $CD = BC - r = 63 - r$ and $AF + CD = AC$ (tangent lengths); $(16 - r) + (63 - r) = 65$ gives $r = 7$ cm
4	**(a)** 35 cm	$6x + 5 = 4x + 15$ gives $x = 5$; $QA = 6(5) + 5 = 35$ cm
5	**(c)** 100°	$\angle APB = 180^\circ - 40^\circ - 40^\circ$; $\angle APB = 100^\circ$
6	**(c)** square	Four right angles and $OA = OB = PA = PB$
7	**(b)** 35 cm	$r^2 = OP^2 - PT^2 = 37^2 - 12^2 = 1225$; $r = 35$ cm
8	**(b)** 94 cm	Perimeter $= PA + AO + OB + BP = 2(12) + 2(35)$; Perimeter $= 94$ cm
9	**(d)** 672 cm²	Area $= 2 \times \frac{1}{2} \times 48 \times 14$; Area $= 672$ cm^2
10	**(b)** 88 cm	Half-chord $= \sqrt{55^2 - 33^2} = 44$ cm; Chord $= 2 \times 44 = 88$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$PA = PB$, so $\angle PAB = \frac{180^\circ - 72^\circ}{2} = 54^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 54^\circ = 36^\circ$ (1).	1+1
12	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 53^2 - 45^2 = 784$ (1); $PT = 28$ cm (1).	1+1
13	$PA = PB$, so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB$; $\angle APB = 180^\circ - 20^\circ - 20^\circ$ (1); $\angle APB = 140^\circ$ (1).	1+1
14	$PA = PB$, so $\angle PAB = \frac{180^\circ - 84^\circ}{2} = 48^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 48^\circ = 42^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tangents from an external point are equal: $QA = QB$ (1); $10x + 3 = 6x + 19$ gives $x = 4$ (1); $QA = 10(4) + 3 = 43$ cm (1).	1+1+1
16	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $25 + 16 = AD + 19$ (1); $AD = 22$ cm (1).	1+1+1
17	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 35 \times 12$ (1); Area $= 420$ cm^2 (1).	1+1+1
	OR: Tangents from an external point are equal: $QA = QB$ (1); $12x + 5 = 9x + 20$ gives $x = 5$ (1); $QA = 12(5) + 5 = 65$ cm (1).	1+1+1
18	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $30 + 13 = AD + 24$ (1); $AD = 19$ cm (1).	1+1+1
	OR: $\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 36 \times 15$ (1); Area $= 540$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{55^2 - 33^2} = 44$ cm; perimeter $= 2(44 + 33)$ (2); Perimeter 154 cm; area $= 2 \times \frac{1}{2} \times 33 \times 44 = 1452$ cm^2 (1).	2+2+1
	OR: Let the chord AB of the larger circle touch the smaller circle at M; join OM (1); $OM \perp AB$ (radius of the smaller circle $\perp$ tangent AB) (2); A perpendicular from the centre of the larger circle to a chord bisects the chord (1); Hence $AM = MB$ (1).	1+2+1+1
20	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{58^2 - 40^2} = 42$ cm; perimeter $= 2(42 + 40)$ (2); Perimeter 164 cm; area $= 2 \times \frac{1}{2} \times 40 \times 42 = 1680$ cm^2 (1).	2+2+1
	OR: $AC = \sqrt{16^2 + 63^2} = 65$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 16 - r$ and $CD = CE = 63 - r$ and $AE + EC = AC$: $(16 - r) + (63 - r) = 65$ (2); $r = 7$ cm, $AF = 9$ cm, $CD = 56$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90^\circ$; the correct sum is 180°.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 cm	$AF = AB - r = 25 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(25 - r) + (60 - r) = 65$ gives $r = 10$ cm
2	**(b)** 56 cm	$9x + 2 = 7x + 14$ gives $x = 6$; $QA = 9(6) + 2 = 56$ cm
3	**(a)** 80°	$\angle APB = 180^\circ - 50^\circ - 50^\circ$; $\angle APB = 80^\circ$
4	**(d)** 60°	$\cos \angle TOP = OT/OP = \frac{1}{2}$
5	**(b)** 36 cm	$r^2 = OP^2 - PT^2 = 39^2 - 15^2 = 1296$; $r = 36$ cm
6	**(d)** 102 cm	Perimeter $= PA + AO + OB + BP = 2(15) + 2(36)$; Perimeter $= 102$ cm
7	**(d)** 1200 cm²	Area $= 2 \times \frac{1}{2} \times 40 \times 30$; Area $= 1200$ cm^2
8	**(a)** 84 cm	Half-chord $= \sqrt{58^2 - 40^2} = 42$ cm; Chord $= 2 \times 42 = 84$ cm
9	**(c)** 12 cm	$AF = AB - r = 33 - r$, $CD = BC - r = 56 - r$ and $AF + CD = AC$ (tangent lengths); $(33 - r) + (56 - r) = 65$ gives $r = 12$ cm
10	**(c)** 56 cm	$11x + 1 = 8x + 16$ gives $x = 5$; $QA = 11(5) + 1 = 56$ cm

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 55^2 - 44^2 = 1089$ (1); $PT = 33$ cm (1).	1+1
12	$PA = PB$, so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB$; $\angle APB = 180^\circ - 70^\circ - 70^\circ$ (1); $\angle APB = 40^\circ$ (1).	1+1
13	$PA = PB$, so $\angle PAB = \frac{180^\circ - 96^\circ}{2} = 42^\circ$; $\angle OAP = 90^\circ$ (radius $\perp$ tangent) (1); $\angle OAB = 90^\circ - 42^\circ = 48^\circ$ (1).	1+1
14	The radius to the point of contact T is perpendicular to the tangent, so $\triangle OTP$ is right-angled at T; $PT^2 = OP^2 - OT^2 = 58^2 - 42^2 = 1600$ (1); $PT = 40$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tangents from an external point are equal: $QA = QB$ (1); $8x + 7 = 5x + 22$ gives $x = 5$ (1); $QA = 8(5) + 7 = 47$ cm (1).	1+1+1
16	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $22 + 20 = AD + 18$ (1); $AD = 24$ cm (1).	1+1+1
17	$\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 32 \times 24$ (1); Area $= 768$ cm^2 (1).	1+1+1
	OR: Tangents from an external point are equal: $QA = QB$ (1); $13x + 6 = 9x + 14$ gives $x = 2$ (1); $QA = 13(2) + 6 = 32$ cm (1).	1+1+1
18	For a quadrilateral circumscribed about a circle, $AB + CD = AD + BC$ (1); $26 + 15 = AD + 20$ (1); $AD = 21$ cm (1).	1+1+1
	OR: $\angle OAP = \angle OBP = 90^\circ$, so $PAOB$ is made of two congruent right triangles (1); Area $= 2 \times \frac{1}{2} \times 40 \times 9$ (1); Area $= 360$ cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{60^2 - 36^2} = 48$ cm; perimeter $= 2(48 + 36)$ (2); Perimeter 168 cm; area $= 2 \times \frac{1}{2} \times 36 \times 48 = 1728$ cm^2 (1).	2+2+1
	OR: $AC = \sqrt{25^2 + 60^2} = 65$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 25 - r$ and $CD = CE = 60 - r$ and $AE + EC = AC$: $(25 - r) + (60 - r) = 65$ (2); $r = 10$ cm, $AF = 15$ cm, $CD = 50$ cm (1).	2+2+1
20	$OA \perp PA$, so $PA^2 = OP^2 - OA^2$ and $PA = PB$ (2); $PA = \sqrt{61^2 - 11^2} = 60$ cm; perimeter $= 2(60 + 11)$ (2); Perimeter 142 cm; area $= 2 \times \frac{1}{2} \times 11 \times 60 = 660$ cm^2 (1).	2+2+1

	OR: $AC = \sqrt{33^2 + 56^2} = 65$ cm; $BDOF$ is a square of side r, so $BF = BD = r$ (2); $AF = AE = 33 - r$ and $CD = CE = 56 - r$ and $AE + EC = AC$: $(33 - r) + (56 - r) = 65$ (2); $r = 12$ cm, $AF = 21$ cm, $CD = 44$ cm (1).	2+2+1

Mistakes to avoid:

- Taking the tangent as perpendicular to the *chord* or parallel to the radius; it is perpendicular to the radius at the contact point.
- Using the Pythagoras theorem with the wrong hypotenuse: in $\triangle OPT$ the hypotenuse is OT (from the centre to the external point).
- Forgetting that the tangent lengths from one external point are equal and stopping after finding only one of them.
- Claiming a line meeting the circle at two points is a tangent; that line is a secant.
- Writing $\angle AOB + \angle APB = 90°$; the correct sum is $180°$.
- Assuming a tangent can be drawn from a point inside the circle.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND